FORWARD DEPLOYED / PROPERTY / SAFETY CRITICAL

MAINTENANCE TRIAGE

10

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

Can a triage system cut routine handling effort while guaranteeing that no safety critical report is delayed or downgraded?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A proposed triage system for a property manager with about 480 units that classifies and links maintenance reports and escalates gas, fire, flood and safeguarding at once.

- Treat escalation as the default path and make every downgrade need explicit evidence.
- Accept email and phone transcripts and app reports in one queue linked to units and contractors.
- Use synthetic reports and units only and never imply a safety rating or live results.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Report intake** → 2 STAGE **Classify and link** → 3 STAGE **Safety escalation** → 4 STAGE **Dispatch owner**

- Intake adapters for email and phone transcripts and app forms normalised into one report record with source.
- Rules plus model classification where a safety keyword or low confidence triggers escalation before other steps.
- Unit and contractor linking with duplicate grouping and a dispatch queue that records which person made each decision.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ A synthetic test cannot certify safety performance and live use would need regulatory review.

⚠ Transcripts can be garbled so the design must assume some reports are misread and escalate on doubt.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

MAINTENANCE TRIAGE / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

10

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EXPERIMENTAL METHODOLOGY

01 Generate synthetic reports with labelled hazards plus vague or garbled phrasing and duplicates and multi unit incidents.

02 Compare keyword rules only and model only and combined fail safe designs on safety recall and false escalation volume.

03 Measure missed safety reports and time to escalation and duplicate accuracy and reviewer workload on held out synthetic sets.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Safety report recall with zero tolerated misses on tests
- Time to escalation under noisy and ambiguous reports
- Duplicate grouping accuracy and routing correctness

REQUIRED REPOSITORY ARTEFACTS

- Synthetic report and unit dataset
- Escalation rule set with safety recall test suite
- Triage queue demo and evaluation report

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.